Collapse Analysis of Entropy–Information Scaling Across Spatial Dimensions

Shared Structure and Dimensional Degeneracy

in Nonlinear Field Systems

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Abstract

The Critical Coherence Index (CCI) has been shown to follow an empirical entropy–information scaling relation of the form

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}$$

in one-, two-, and three-dimensional ϕ^4 field dynamics [2, 3, 5]. While individual dimensional studies established strong monotonic correlations, the question of whether a unified cross-dimensional description exists remained open.

In this work we perform a direct collapse analysis by plotting the CCI against the best scaling ratio R_α for all three dimensions simultaneously, using log-x and log-log representations.

We find a partial collapse: a shared monotonic structure is visible across dimensions, but no clean data collapse onto a single curve is achieved. The 3D data show increased dispersion, consistent with the previously identified exponent degeneracy.

We interpret this as a projection effect: in low dimensions the system admits a single effective exponent, while in higher dimensions multiple competing reorganisation channels lead to a degeneracy of effective descriptions. The exponent plateau in 3D reflects a transition to a higher-dimensional scaling manifold rather than a failure of the scaling principle.

All results are empirical; a theoretical derivation remains open.

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1 Introduction

Previous work in this series established the entropy–information scaling of the CCI across spatial dimensions:

- 1D: $\alpha \approx 2.5$, $r_s \approx 0.73$ [2]
- 2D: $\alpha \gtrsim 3.0$, $r_s \approx 0.87$ [3]
- 3D: plateau $\alpha \in [2.8, 3.5]$, $r_s \approx 0.85$ [5]

While the scaling persists in all dimensions, no unified cross-dimensional picture has yet been established. The goal of this paper is to test whether the data from all three dimensions can be collapsed onto a shared functional form.

2 Method

For each dimension d, we compute the scaling ratio:

$$R_\alpha^{(d)} = \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{(I_{\mathrm{nn}} + \varepsilon)^{\alpha_d}}, \tag{1}$$

using the representative effective exponents: $\alpha_{1D} = 2.5$, $\alpha_{2D} = 3.0$, $\alpha_{3D} = 2.8$ (representative point from the plateau).

The collapse is visualised in log-x and log-log representations. A normalised version (dividing by the ensemble mean) provides an additional cross-dimensional comparison.

All data are drawn from the ensemble CSV files produced by the upstream simulation scripts [6].

3 Results

3.1 Partial Collapse

Figure 1 shows the collapse attempt across dimensions.

A shared monotonic trend is visible across all three dimensions in both representations. However, the data do not collapse onto a single curve:

- the 1D and 2D clouds show a tighter structure,
- the 3D data show increased dispersion,
- the ranges of R_α differ across dimensions.

3.2 Summary Table

Table 1: Cross-dimensional scaling properties used in the collapse.

Dimension	α used	r_s	R^2	Structure
1D	2.5	0.73	moderate	sharp exponent
2D	3.0	0.87	0.74	shifted exponent
3D	2.8	0.85	0.37	plateau [2.8, 3.5]

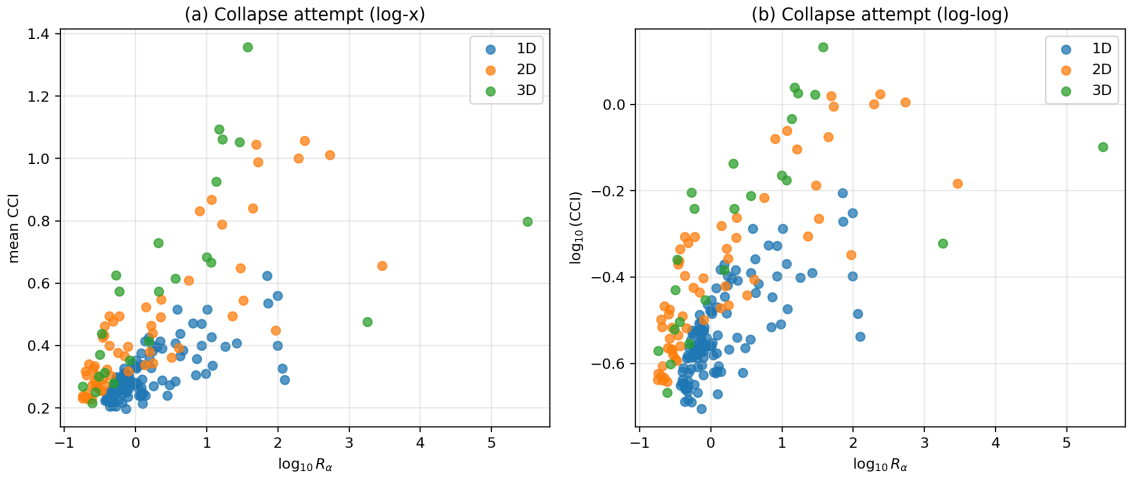

Figure 1: **Collapse attempt across dimensions.** *(a)* Log-x representation of $\mathcal{C}_{\mathrm{CCI}}$ vs R_α showing a shared monotonic structure across 1D, 2D, and 3D. *(b)* Log-log representation indicating a similar functional trend with increasing dispersion in 3D. A partial collapse is observed: the monotonic structure is shared, but no single curve describes all dimensions.

4 Discussion

4.1 Interpretation as a Projection Effect

The partial collapse supports the interpretation that the entropy–information scaling relation captures a shared structural dependency across dimensions. In low dimensions, the system is effectively constrained and admits a single effective exponent; in higher dimensions, multiple competing channels of structural reorganisation lead to a degeneracy of effective exponents.

Within the broader framework [1], the entropy–information ratio $\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ acts as an empirical projection of the structural free energy onto a lower-dimensional scaling plane. The exponent plateau in 3D reflects an increasing number of relevant projection directions, making a scalar exponent insufficient.

4.2 Conceptual Picture

- **1D:** clear exponent, tight collapse
- **2D:** shifted exponent, strong collapse
- **3D:** degenerate exponent, partial collapse

The transition from a well-defined exponent to an exponent plateau is a transition to a higher-dimensional scaling manifold, not a failure of the scaling principle.

5 Conclusion

We have performed a direct cross-dimensional collapse analysis of the entropy–information scaling of the CCI.

A partial collapse is observed: the monotonic structure is shared across 1D, 2D, and 3D, but no single curve provides a universal description.

Key result:

Scaling survives, but the exponent does not.
The partial collapse reflects a shared structural principle
expressed through dimension-dependent effective parameters.

Future work will investigate multi-parameter scaling descriptions and seek a theoretical derivation of the cross-dimensional structure.

All results are empirical. A first-principles derivation of the cross-dimensional scaling structure remains an open problem.

A Reproducibility

All code, data, and figures are available at:

https://github.com/Chris4081/structural-selection-principle

Step 1 — Install dependencies:

```
pip install numpy pandas matplotlib scipy scikit-learn
```

Step 2 — Generate the required CSV data files:

```
# 1D ensemble  →  cci_entropy_information_test.csv
python cci_alpha_scaling.py

# 2D ensemble  →  2d_cci_entropy_information_test.csv
python cci_entropy_scaling_2d.py

# 3D ensemble  →  3d_cci_entropy_information_test.csv
python cci_entropy_scaling_3d.py
```

Step 3 — Generate the collapse figures:

```
python Collapse-Plot.py
```

Output files generated:

- `collapse_plot_summary.png` — Figure 1: log-x and log-log side-by-side
- `collapse_plot_linear.png` — linear scale version
- `collapse_plot_logx.png` — log-x only
- `collapse_plot_loglog.png` — log-log only
- `collapse_plot_normalized.png` — mean-normalised cross-dimensional comparison

Key parameters used:
$\alpha_{1D} = 2.5$, $\alpha_{2D} = 3.0$, $\alpha_{3D} = 2.8$ (representative plateau point).

References

[1] C. Krieg, *A Structural Free Energy on the Solution Manifold*, preprint, available from the author (2026).

[2] C. Krieg, *Entropy–Information Scaling of the Critical Coherence Index in Nonlinear Field Dynamics*, preprint, available from the author (2026).

[3] C. Krieg, *Entropy–Information Scaling of the Critical Coherence Index in Two-Dimensional ϕ^4 Field Dynamics*, preprint, available from the author (2026).

[4] C. Krieg, *Dimensional Dependence of Entropy–Information Scaling in Nonlinear Field Systems*, preprint, available from the author (2026).

[5] C. Krieg, *Persistence of Entropy–Information Scaling in Three Dimensions*, preprint, available from the author (2026).

[6] C. Krieg, *Structural Selection — companion code*, https://github.com/Chris4081/structural-selection-principle (2026).